Statistical Evaluation of Hypotheses in Machine Learning

Comprehensive Summary and Numerical Analysis

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1 Introduction

Evaluating the performance of a hypothesis h is critical to understanding its generalization power. This document covers error definitions, estimation problems, confidence intervals, and comparative statistical tests.

2 Two Definitions of Error

2.1 True Error

The **true error** of hypothesis h with respect to target function f and distribution $\mathcal{D}$ is the probability that h will misclassify an instance drawn at random according to $\mathcal{D}$.

$$error_{\mathcal{D}}(h) \equiv \Pr_{x \in \mathcal{D}}[f(x) \neq h(x)] \tag{1}$$

2.2 Sample Error

The **sample error** of h with respect to target function f and data sample S is the proportion of examples h misclassifies.

$$error_S(h) \equiv \frac{1}{n} \sum_{x \in S} \delta(f(x) \neq h(x)) \tag{2}$$

Where $\delta = 1$ if $f(x) \neq h(x)$, and 0 otherwise.

Numerical Example: In a test set S of $n = 40$ examples, hypothesis h misclassifies 12.

$$error_S(h) = \frac{12}{40} = 0.30$$

3 Distributions and Central Limit Theorem

3.1 Binomial Distribution: The Exact Model

The probability of observing exactly r misclassifications in n independent trials is governed by the Binomial distribution. This is the discrete foundation of error estimation.

$$P(r) = \frac{n!}{r!(n-r)!} p^r (1-p)^{n-r} \tag{3}$$

- **Expected Errors:** $E[X] = np$ (The average number of errors we expect).
- **Standard Deviation:** $\sigma = \sqrt{np(1-p)}$ (The expected spread/fluctuation of errors).

3.2 Normal Distribution: The Practical Approximation

For large n (typically $n \geq 30$ and $np(1-p) \geq 5$), the discrete Binomial distribution is approximated by the continuous Normal distribution:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} \tag{4}$$

This approximation is significant because it enables the use of standard critical values (z_N) to define confidence intervals.

3.3 Central Limit Theorem: The Theoretical Bridge

The **Central Limit Theorem** states that as $n \to \infty$, the distribution of the sample mean $\bar{Y}$ approaches a Normal distribution, even if the underlying distribution is not normal.

- **Mean:** μ
- **Standard Error:** $\sigma_{\bar{Y}} = \sigma/\sqrt{n}$

This theorem allows us to treat the sample error $error_S(h)$ as a random variable drawn from a Normal distribution centered at the true error $error_{\mathcal{D}}(h)$.

Numerical Example: Suppose a model has a true error rate $p = 0.30$. If we test it on $n = 40$ examples:

1. The expected number of errors is $40 \times 0.3 = 12$.
2. The standard deviation of the error proportion is:

$$\sigma_{error_S} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.3(0.7)}{40}} = \sqrt{0.00525} \approx 0.0725$$

3. Using the Normal approximation, we can say that approximately 95% of the time, our observed sample error will fall within $0.30 \pm (1.96 \times 0.0725)$, or between 15.8% and 44.2%.

4 Problems Estimating Error: Bias and Variance

4.1 Bias

If S is the training set, $error_S(h)$ is an **optimistically biased** estimate. For an unbiased estimate, h and S must be independent.

$$bias \equiv E[error_S(h)] - error_{\mathcal{D}}(h) \tag{5}$$

4.2 Variance

Even with an unbiased S, $error_S(h)$ may vary from $error_{\mathcal{D}}(h)$ depending on the specific sample chosen.

Numerical Example of Bias and Variance: Suppose the True Error of a model is $error_{\mathcal{D}}(h) = 0.20$.

- **Bias Calculation:** We test the model on its own training data and observe a sample error of 0.05. The Bias is $0.05 - 0.20 = -0.15$ (Optimistic bias).
- **Variance Calculation:** We test the model on three different independent test sets ($n = 100$ each) and observe errors of $0.18, 0.22$, and 0.20. The variance of these observations is:

$$\text{Mean} = 0.20, \quad Var = \frac{(0.18 - 0.2)^2 + (0.22 - 0.2)^2 + (0.2 - 0.2)^2}{3 - 1} = 0.0004$$

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5 Confidence Intervals

If $n \geq 30$ and examples are independent, we can conclude that with approximately $N\%$ probability, $error_{\mathcal{D}}(h)$ lies in the interval:

$$error_S(h) \pm z_N \sqrt{\frac{error_S(h)(1 - error_S(h))}{n}} \tag{6}$$

$N\%$:	50%	68%	80%	90%	95%	98%	99%
z_N:	0.67	1.00	1.28	1.64	1.96	2.33	2.58

Numerical Example: For $n = 100$ and $error_S(h) = 0.2$, the 95% confidence interval ($z_N = 1.96$) is:

$$0.2 \pm 1.96\sqrt{\frac{0.2(0.8)}{100}} = 0.2 \pm 1.96(0.04) = 0.2 \pm 0.078$$

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6 Comparing Learning Methods

6.1 Difference Between Hypotheses

To estimate $d \equiv error_{\mathcal{D}}(h_1) - error_{\mathcal{D}}(h_2)$, we use:

$$\hat{d} \pm z_N \sqrt{\frac{error_{S_1}(h_1)(1 - error_{S_1}(h_1))}{n_1} + \frac{error_{S_2}(h_2)(1 - error_{S_2}(h_2))}{n_2}} \tag{7}$$

Numerical Example: $error_{S1}(h_1) = 0.2$, $error_{S2}(h_2) = 0.25$, $n_1 = n_2 = 100$. $\hat{d} = -0.05$. Standard Error $= \sqrt{\frac{0.16}{100} + \frac{0.1875}{100}} \approx 0.059$. 95% Interval: $-0.05 \pm 1.96(0.059) = [-0.165, 0.065]$.

7 Paired t-Test Formula

When comparing hypotheses h_A and h_B on k disjoint test sets (k-fold CV):

$$\bar{\delta} \pm t_{N,k-1} \sqrt{\frac{1}{k(k-1)} \sum_{i=1}^{k} (\delta_i - \bar{\delta})^2} \tag{8}$$

Numerical Example: $k = 5$ folds with error differences: $\delta = \{0.02, 0.05, 0.01, 0.04, 0.03\}$.

- Mean difference $\bar{\delta} = 0.03$.
- Variance of difference $Var(\delta) \approx 0.00025$.
- Standard Error $s_{\bar{\delta}} = \sqrt{\frac{0.00025}{5}} \approx 0.007$.
- 95% Interval ($t_{2.776}$): $0.03 \pm 2.776(0.007) \approx [0.01, 0.05]$.

Numerical Example: With $k = 5$ folds, differences δ_i are $[0.01, 0.02, 0.01, 0.03, 0.02]$. $\bar{\delta} = 0.018$. $s_{\bar{\delta}} = \sqrt{\frac{0.00028}{20}} \approx 0.0037$. Using $t_{95\%,4} = 2.776$, the interval is 0.018 ± 0.010.

8 Significance Testing (One-Tailed t-Test)

To verify if Algorithm B is strictly better than Algorithm A (i.e., error difference > 0), we use a one-tailed t-test.

8.1 Hypotheses

- **Null (H_0):** Mean difference $\mu_\delta \leq 0$ (No improvement).
- **Alternative (H_a):** Mean difference $\mu_\delta > 0$ (Improvement).

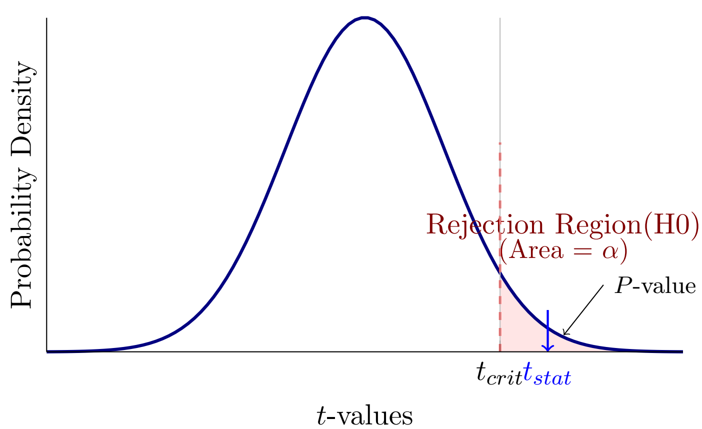
8.2 The t-Statistic

The t-statistic measures how far the sample mean deviates from the null hypothesis, in units of standard error.

$$t = \frac{\bar{\delta}}{s/\sqrt{n}} \tag{9}$$

Note: We divide by $\sqrt{n}$ to convert standard deviation (s) into standard error.

8.3 Visualizing the Test



8.4 Numerical Example

We compare two models on $n = 25$ folds. The mean accuracy improvement is $\bar{\delta} = 2.0\%$ with standard deviation $s = 4.0\%$.

1. **Calculate t-statistic:**
$$t = \frac{2.0}{4.0/\sqrt{25}} = \frac{2.0}{4.0/5} = \frac{2.0}{0.8} = 2.5$$

2. **Find Critical Value:** For $df = 24$ and $\alpha = 0.05$ (one-tail), $t_{crit} \approx 1.71$.

3. **Decision:** Since $t_{stat}(2.5) > t_{crit}(1.71)$, the result falls in the **Rejection Region**. The P-value is the area to the right of 2.5 (approx 0.009). We reject H_0 and conclude the improvement is significant.